

## 3968. Maximum Manhattan Distance After All Moves

Solved ●

Medium  Topics  Hint

You are given a string `moves` consisting of the characters 'U', 'D', 'L', 'R', and '\_'.

Starting from the origin  $(0, 0)$ , each character represents one move on a 2D plane:

- 'U': Move up by 1 unit.
- 'D': Move down by 1 unit.
- 'L': Move left by 1 unit.
- 'R': Move right by 1 unit.
- '\_': Can be independently replaced with any one of 'U', 'D', 'L', or 'R'.

Return the maximum **Manhattan distance** from the origin that can be achieved after all moves have been performed.

### Example 1:

**Input:** `moves = "L_D_"`

**Output:** 4

#### Explanation:

One optimal choice is:

- 'L':  $(0, 0) \rightarrow (-1, 0)$
- '\_' treated as 'D':  $(-1, 0) \rightarrow (-1, -1)$
- 'D':  $(-1, -1) \rightarrow (-1, -2)$
- '\_' treated as 'L':  $(-1, -2) \rightarrow (-2, -2)$

The final Manhattan distance from the origin is  $|0 - (-2)| + |0 - (-2)| = 4$ .

### Example 2:

**Input:** `moves = "U_R"`

**Output:** 3

#### Explanation:

One optimal choice is:

- 'U':  $(0, 0) \rightarrow (0, 1)$
- '\_' treated as 'U':  $(0, 1) \rightarrow (0, 2)$
- 'R':  $(0, 2) \rightarrow (1, 2)$

The final Manhattan distance from the origin is  $|0 - 1| + |0 - 2| = 3$ .

### Constraints:

- $1 \leq \text{moves.length} \leq 10^5$
- `moves` consists of only 'U', 'D', 'L', 'R', and '\_'.

Seen this question in a real interview before? 1/6

Yes No

Accepted 44,663/70.4K | Acceptance Rate 63.4%

 Topics



 Hint 1

 Hint 2

-  Hint 3 
- 
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